PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS-C702

Carbomer

INCI	CAS	EC	FAMILY	SUPPLIED AS
Carbomer	9007-20-9	—	PC-03	Powder

01 POSITIONING

Four grades, one rheology language — pick the curve, we ship the powder that draws it.

02 HOW IT WORKS

01 Swell on neutralization
Neutralized carboxyls repel, swelling each microgel up to 1000× — viscosity from 0.2% dosage.

02 Yield stress
Suspends beads, pigments and bubbles indefinitely, yet flows instantly under shear.

03 Crystal clarity
Fully hydrated gels transmit light — the transparent-gel category exists on this polymer.

03 APPLICATIONS

→ Hydroalcoholic and moisturizing gels
→ Bead-suspension shower gels
→ Styling gels
→ Toothpastes (grade-dependent)
→ Emulsion stabilization

04 FORMULATION GUIDE

USE LEVEL	0.2–1%
PH & CONDITIONS	Neutralize to pH 5.5–7 (NaOH, TEA, arginine); viscosity collapses below pH 4 and above 10

— Disperse slowly into vortexing water before neutralizing; hydrate fully to avoid fish-eyes.

— Electrolytes thin carbomer gels — for salt-rich systems choose 702/500-type grades.

05 TYPICAL SPECIFICATION

Appearance	White fluffy powder
Viscosity (0.5%, neutralized)	per grade: 40 000–60 000 mPa·s (940-type)
Loss on drying	≤ 2%
Residual benzene	≤ 2 ppm (per TDS)

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Viscosity per 0.5% dosage
Carbomer 940-type

Suspension of 1 mm beads (30 d)
Carbomer gel

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.